

LOLR Model and Sunspot Results

April 23, 2026

- State variables in period t : private debt b_t , LOLR debt l_t , growth state $g_t \in \{g_L, g_H\}$, and transitory shock ε_t .
- Output in repayment: $y_t = g_t \exp(\sigma_\varepsilon \varepsilon_t)$.
- Output in default: $y_t^D = \phi_{g_t} y_t$ with state-dependent output loss.
- Growth follows a Markov chain P_g ; ε_t is iid with invariant probabilities π_ε .
- International lenders are risk-neutral with gross risk-free rate $1 + r^*$.

Government Problem in Repayment

Given current (b, l, g, ε) and no default, the sovereign chooses next debt (b', l') :

$$V^{ND}(b, l, g, \varepsilon) = \max_{b', l'} \{ u(c) + \beta g^{1-\gamma} \mathbb{E} [\max \{ V^{ND}(b', l', g', \varepsilon'), V^D(b', l', g', \varepsilon') \}] \}.$$

Consumption in repayment is

$$c = y - b - l + n(b', l', g) + n_l(l', g),$$

where

$$n(b', l', g) = \frac{g b'}{R(b', l', g)}, \quad n_l(l', g) = \frac{g l'}{R_l^{ND}}.$$

Government Problem in Default

In default, private market issuance is shut down and the sovereign only chooses LOLR debt:

$$V^D(b, l, g, \varepsilon) = \max_{l'} \left\{ u \left(y^D - l + \frac{g l'}{R_l^D} \right) + \beta g^{1-\gamma} \mathbb{E} \left[\theta V^D(\tilde{b}, l', g', \varepsilon') \right. \right. \\ \left. \left. + (1 - \theta) \max \{ V^{ND}(\kappa \tilde{b}, l', g', \varepsilon'), V^D(\tilde{b}, l', g', \varepsilon') \} \right] \right\},$$

with transformed debt states

$$\tilde{b} = \frac{b}{g}, \quad \kappa \tilde{b} \text{ after restructuring on re-entry.}$$

How Q and X Are Built

At current state $(b, \ell, g, \varepsilon)$:

$$d(b, \ell, g, \varepsilon) = \mathbf{1}\{V^{ND}(b, \ell, g, \varepsilon) < V^D(b, \ell, g, \varepsilon)\}.$$

Bond price is the payoff decomposition

$$Q(b, \ell, g, \varepsilon) = (1 - d) + d X(b, \ell, g, \varepsilon),$$

where:

- if no default ($d = 0$), lenders receive 1;
- if default ($d = 1$), lenders receive continuation value X .

Continuation value satisfies

$$X(b, \ell, g, \varepsilon) = \frac{1}{1 + r^*} \mathbb{E}[\theta X' + (1 - \theta)((1 - e')X' + e' \kappa Q') \mid g],$$

with $e' = \mathbf{1}\{V^{ND}(\kappa b', \ell', g', \varepsilon') \geq V^D(b', \ell', g', \varepsilon')\}$.

So in the code, (Q, X) are obtained jointly as a fixed point given current policy rules and the default/re-entry decisions.

Pricing, Constraints, and LOLR Policy

- Debt price recursion implies $Q = (1 - d) + dX$, with default indicator $d = \mathbf{1}\{V^{ND} < V^D\}$.
- Gross market schedule and implied private issuance:

$$R(b', l', g) = \frac{1 + r^*}{\mathbb{E}[Q]}, \quad n(b', l', g) = \frac{g b'}{R(b', l', g)}.$$

- Endogenous default probability schedule:

$$p^{def}(b', l', g) = \mathbb{E}[d'].$$

- Feasible repayment choices must satisfy:

$$p^{def}(b', l', g) \leq \bar{p}_u, \quad n(b', l', g) \leq \bar{n}_g, \quad l' \leq \bar{l}^{ND}.$$

- In default, LOLR borrowing is capped by $l' \leq \bar{l}^D$.

Numerical Solution in the Code

The solver in `1period_lolr/src/solver_lolr.jl` iterates:

- Solve default value V^D and default-state LOLR policy l'_D .
- Update default decision d and re-entry decision $e = 1 - d$.
- Solve bond-price fixed point for continuation value X and price Q .
- Compute schedules (R, n, p^{def}) .
- Update repayment value V^{ND} under feasibility constraints.
- Recover optimal policy indexes (b', l') for simulation.

Simulation in `simulation_lolr.jl` then generates moments, default episodes, and debt distributions.

Sunspot Outputs Used in This Section

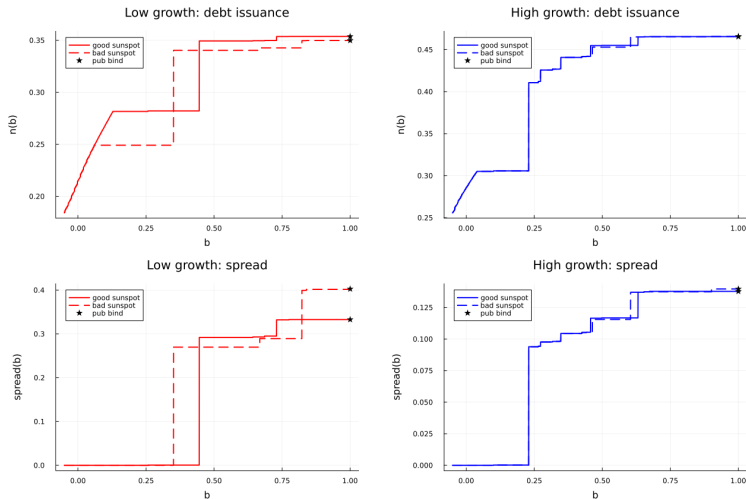
- Moments table file: `../1period_sunspot/result/sunspot_moments_pb.tex`.
- Policy figure file: `../1period_sunspot/result/sunspot_policy_2x2.png`.
- The table compares $P_b = 25\%$ vs. $P_b = 1\%$ for:
 - First moments.
 - Low-growth-state moments.
 - High-growth-state moments.

Sunspot Moments Table

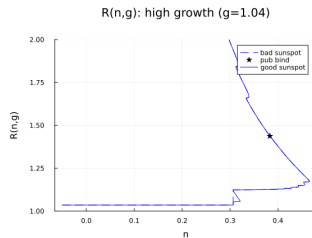
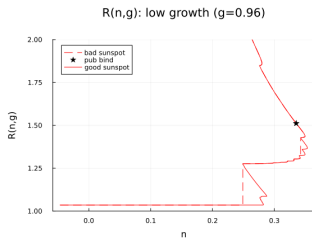
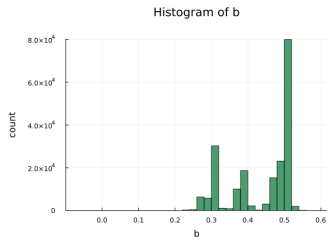
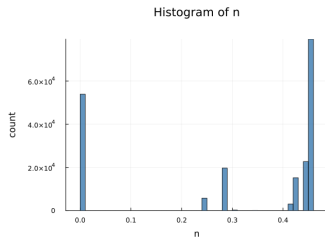
Table: Sunspot moments for alternative bad-sunspot probabilities (P_b)

Moment	P_b=25%	P_b=1%
<i>First moments</i>		
avg(spread)	0.0908	0.0013
avg(qb/y)	0.4036	0.3389
avg(f/y)	0.4569	0.3512
avg(n/y)	0.4036	0.3389
avg(b/y)	0.4116	0.3482
avg(tb/y)	0.0080	0.0092
default rate	0.1526	0.0042
<i>Low-growth state</i>		
avg(spread)	0.0007	0.0020
avg(qb/y)	0.2858	0.3388
avg(f/y)	0.2960	0.3514
avg(n/y)	0.2858	0.3388
avg(b/y)	0.3093	0.3654
avg(tb/y)	0.0235	0.0266
default rate	0.3865	0.0109
<i>High-growth state</i>		
avg(spread)	0.1110	0.0009
avg(qb/y)	0.4300	0.3390
avg(f/y)	0.4929	0.3512
avg(n/y)	0.4300	0.3390
avg(b/y)	0.4346	0.3375
avg(tb/y)	0.0046	-0.0015
default rate	0.0000	0.0000

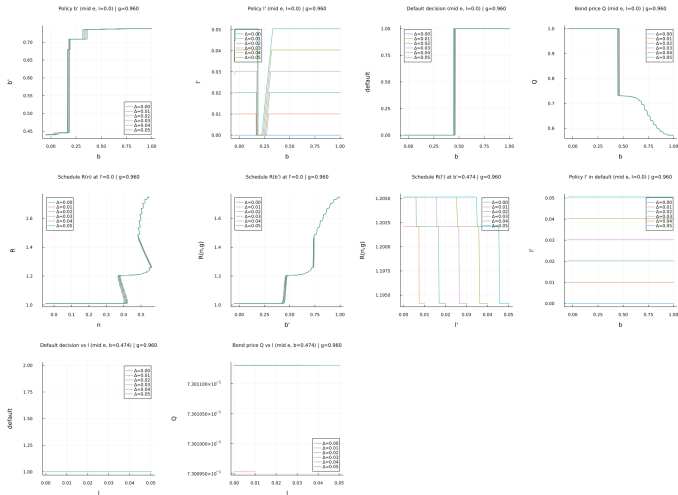
Sunspot Optimal Policy: 2x2 Figure



Sunspot n and R Figure



LOLR Policies by Δ (Spread 0.06)



LOLR Moments by Δ at Spread 0.06

Table: LOLR moments across Δ at fixed spread = 0.06

Moment	$\Delta=0.00$	$\Delta=0.01$	$\Delta=0.02$	$\Delta=0.03$	$\Delta=0.04$	$\Delta=0.05$
<i>First moments</i>						
avg(spread)	0.0921	0.0923	0.0921	0.0921	0.0921	0.0921
avg(qb/y)	0.6701	0.6701	0.6701	0.6701	0.6701	0.6701
avg(f/y)	0.7386	0.7386	0.7386	0.7386	0.7386	0.7386
avg(n/y)	0.6701	0.6701	0.6701	0.6701	0.6701	0.6701
avg(b/y)	0.6786	0.6786	0.6786	0.6786	0.6786	0.6786
avg(tb/y)	0.0085	0.0088	0.0091	0.0093	0.0096	0.0099
default rate	0.1480	0.1480	0.1480	0.1480	0.1480	0.1480
<i>Low-growth state</i>						
avg(spread)	0.2510	0.2510	0.2510	0.2510	0.2510	0.2510
avg(qb/y)	0.5833	0.5832	0.5833	0.5833	0.5833	0.5833
avg(f/y)	0.7355	0.7353	0.7355	0.7355	0.7355	0.7355
avg(n/y)	0.5833	0.5832	0.5833	0.5833	0.5833	0.5833
avg(b/y)	0.4220	0.4220	0.4220	0.4220	0.4220	0.4220
avg(tb/y)	-0.1614	-0.1601	-0.1592	-0.1581	-0.1571	-0.1560
default rate	0.3880	0.3880	0.3880	0.3880	0.3880	0.3880
<i>High-growth state</i>						
avg(spread)	0.0921	0.0922	0.0921	0.0921	0.0921	0.0921
avg(qb/y)	0.6702	0.6701	0.6702	0.6702	0.6702	0.6702
avg(f/y)	0.7386	0.7386	0.7386	0.7386	0.7386	0.7386
avg(n/y)	0.6702	0.6701	0.6702	0.6702	0.6702	0.6702
avg(b/y)	0.6788	0.6788	0.6788	0.6788	0.6788	0.6788
avg(tb/y)	0.0086	0.0089	0.0091	0.0094	0.0097	0.0099
default rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000